

Custom Final Project Guidelines

Due May 29 at 10 p.m. (May 27 for graduating students)

Custom final projects should be submitted by email to the instructor (marshalljean@uchicago.edu).

You will assemble a 10-20 page (double-spaced) report delving into the research question described in your proposal. You must submit an individual report and use the HLM software to conduct analyses. Given the wide range of research questions you are all working on, you have considerable latitude in how you arrange this report, and the analyses you present. However, at a minimum, you should include the following:

1. Your research question defined early, and as clearly and concisely as possible.
2. A brief (1 – 1½ page) description of the motivation behind and context surrounding this problem. This can include references to key literature, but is not a literature review.
3. A description of the data. Describe the sample and the population it is drawn from. Define the nesting structure. Describe how key covariates are measured or operationalized. Provide a summary table of descriptive statistics.
4. Write out your primary model in leveled notation (e.g. not mixed). This should be what you think is the best model for examining your research question – the model you have settled on and want to present to me. Define all fixed and random effect coefficients in the model. Tell me why you selected this as your best model. Interpret the coefficients and random effects.
5. If your primary model uses a continuous outcome, check the assumptions underlying the model (linearity, multivariate normality of errors, homoskedasticity) and report on how well they hold up.

If your primary model uses a categorical outcome, describe your main effect of interest in terms of the substantive change in probability of your outcome associated with your treatment (independent variable of interest).
6. Provide a useful graphical representation of key results. What this looks like will depend on your research question and results, and what you want to highlight. Choose something that will help me visualize the meaning behind the numbers you've presented. Explain what the graphic depicts.
7. Provide two additional models that you also considered before settling on your primary model. Interpret the results of each (you do not need to define every term). These models can involve different combinations of covariates or random effects, alternative nesting structures, or alternative outcomes, depending on your project. Explain how the results of these models differ or align with your primary model.
8. Summarize what you have learned from your analyses in substantive terms.